

Hitec University Taxila, Department of Computer Engineering

Lab Manual for Mobile Application Development

Lab-1

Development Environment and Gradle

Table of Contents

Hitec University Taxila, Department of Computer Engineering.....	1
Lab Manual for Mobile Application Development	1
1. Introduction.....	1
2. Activity Time boxing.....	1
3. Objectives of the experiment	1
4. Concept Map	2
5. Gradle	2
6. Tools.....	2
7. Homework before Lab	2
8. Walk through Task.....	3
8.1. System Requirements	3
8.2. Installing the Java Development Kit (JDK).....	3
8.2.1. Windows JDK Installation	3
8.3. Installing Android Studio	3
11. Running Your First App	22
11.1. There are two ways to run the app	22
12. Practice Tasks	26

1. Introduction

The goal of this course is to teach the skills necessary to develop Android based applications using the Android Studio Integrated Development Environment (IDE) and the Android Software Development Kit (SDK).

Beginning with the basics, this course provides an outline of the steps necessary to set up an Android development and testing environment. An overview of Android Studio is included covering areas such as tool windows, the code editor and the Designer tool. An introduction to the architecture of Android is followed by an in-depth look at the design of Android applications and user interfaces using the Android Studio environment. More advanced topics such as database management, content providers and intents are also covered, as are touch screen handling, camera access.

Lab manuals of this course also cover advanced features of Android Studio such as Gradle build configuration and the implementation of build variants to target multiple Android device types from a single project code base.

Assuming you already have some Java programming experience, are ready to download Android Studio and the Android SDK, have access to a Windows, Mac or Linux system and ideas for some apps to develop, you are ready to get started.

2. Activity Time boxing

Table 1: Activity Time Boxing

Task No.	Activity Name	Activity time	Total Time
1	Introduction	20mins	20min
2	Walk Through Task	120mins	120min
3	Evaluation Task	40min	40min

3. Objectives of the experiment

- To get basic understanding regarding Android App Development

- To build a mobile app in android studio
- To understand the Android Application Environment
- To Development Environment on machine

4. Concept Map

Before any work can begin on the development of an Android application, the first step is to configure a computer system to act as the development platform. This involves a number of steps consisting of installing the Java Development Kit (JDK) and the Android Studio Integrated Development Environment (IDE) which also includes the Android Software Development Kit (SDK).

This lab will cover the steps necessary to install the requisite components for Android application development on Windows.

5. Gradle

Gradle is an open-source build automation system that builds upon the concepts of Apache Ant and Apache Maven and introduces a Groovy-based domain-specific language (DSL) instead of the XML form used by Apache Maven for declaring the project configuration.

Maven is a build automation tool used primarily for Java projects. Maven addresses two aspects of building software: first, it describes how software is built and second, it describes its dependencies.

6. Tools

Android Studio 3.3

7. Homework before Lab

Bring your laptop in 2nd lab of MAD.

8. Walk through Task

8.1. System Requirements

Android application development may be performed on any of the following system types:

- Windows 7 (32-bit or 64-bit)
- Windows 8 / Windows 8.1
- Windows 10
- Minimum of 4GB of RAM (6GB is preferred)
- 10GB of available disk space

8.2. Installing the Java Development Kit (JDK)

The Android SDK was developed using the Java programming language. Similarly, Android applications are also developed using Java. As a result, the Java Development Kit (JDK) is the first component that must be installed.

Android development requires the installation of either version 6 or 7 of the Standard Edition of the Java Platform Development Kit. Java is provided in both development (JDK) and runtime (JRE) packages. For the purposes of Android development, the JDK must be installed.

8.2.1. Windows JDK Installation

For Windows systems, the JDK may be obtained from Oracle Corporation's website using the following

URL:

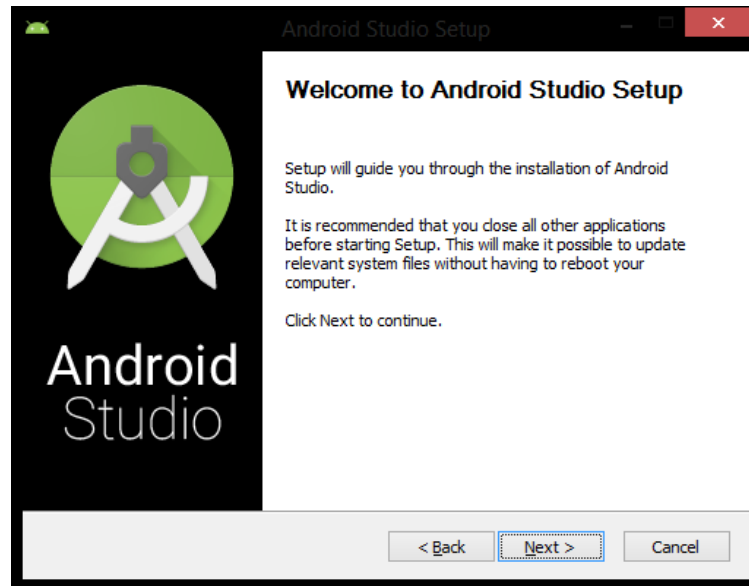
<http://www.oracle.com/technetwork/java/javase/downloads/index.html>

Assuming that a suitable JDK is not already installed on your system, download the latest JDK package that matches the destination computer system. Once downloaded, launch the installation executable and follow the on screen instructions to complete the installation process.

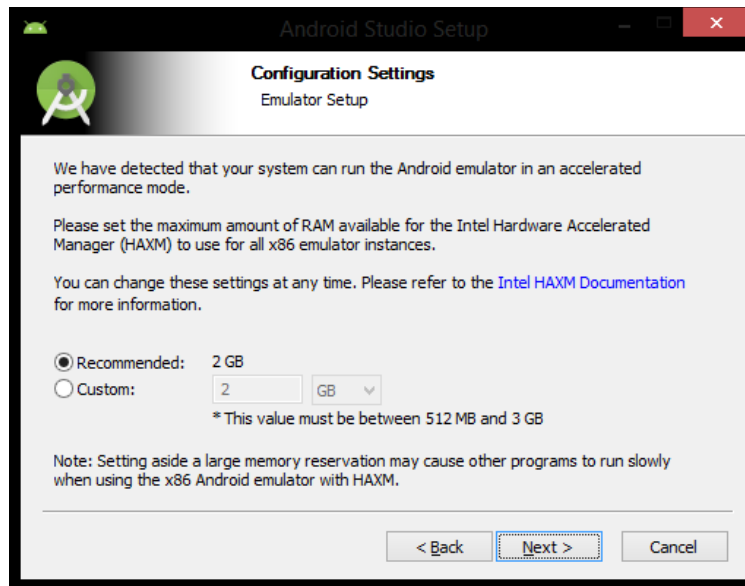
8.3. Installing Android Studio

In this part you will download and install Android Studio which will be the Integrated Development Environment (IDE) used for this course.

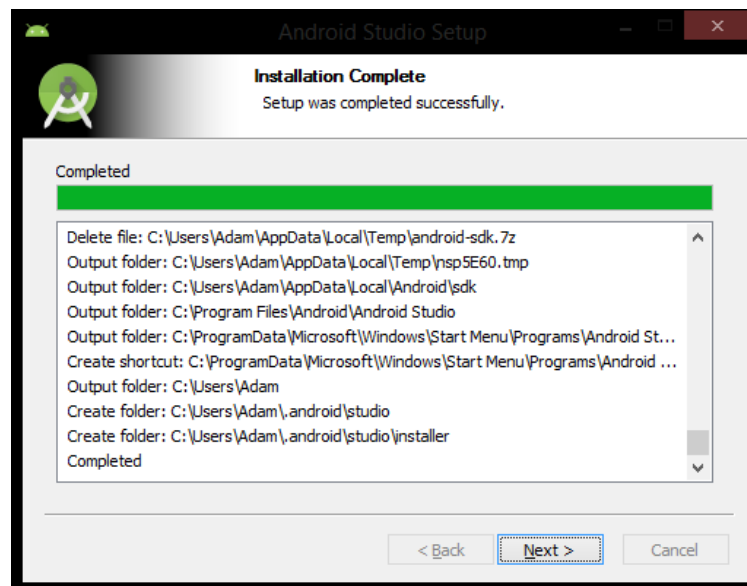
1. Download Android Studio from <http://developer.android.com/sdk/index.html>.
2. Click on 'Download Android Studio'.
3. Open the executable file android-studio-<xxx>.
4. Once the setup loads, you will see the Welcome Screen.



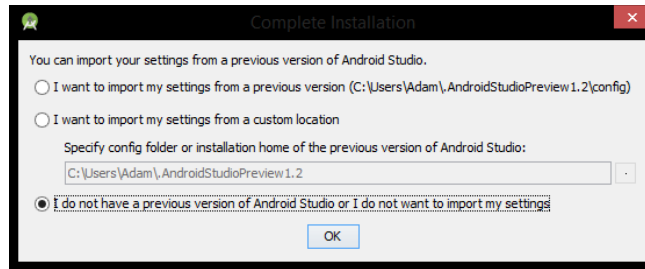
5. Click 'Next >' on the Welcome Screen.
6. When choosing components, ensure all of the checkboxes are checked in for each component to install. Once you are done, click 'Next >'.
7. Agree to the Android Studio and the Intel HAXM License Agreements after reading them.
8. Verify the install locations meet the installation requirements and click 'Next >'.
9. You may or may not see the emulator setup settings, just click 'Next >' after selecting the RAM size.



10. Finally, click 'Install'. You will see which operations are currently running in the installation process and a progress bar displaying their progress.
11. Once the installation process is finished click 'Next >'.



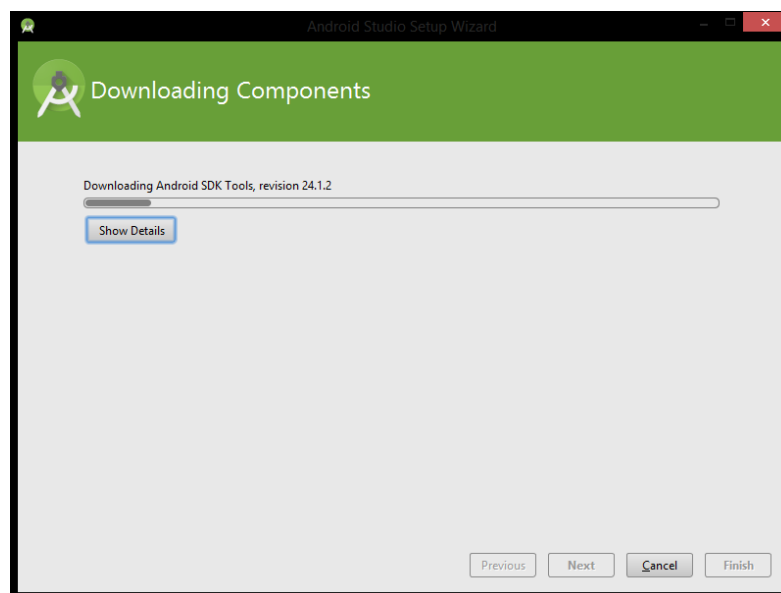
12. Android Studio is now set up. Check on 'Start Android Studio' and click 'Finish'.
13. You will see the Complete Installation screen below.
14. If you had a previous version of Android Studio installed prior, then check either the first of second radio box. Otherwise, check the last radio box and hit 'OK'.



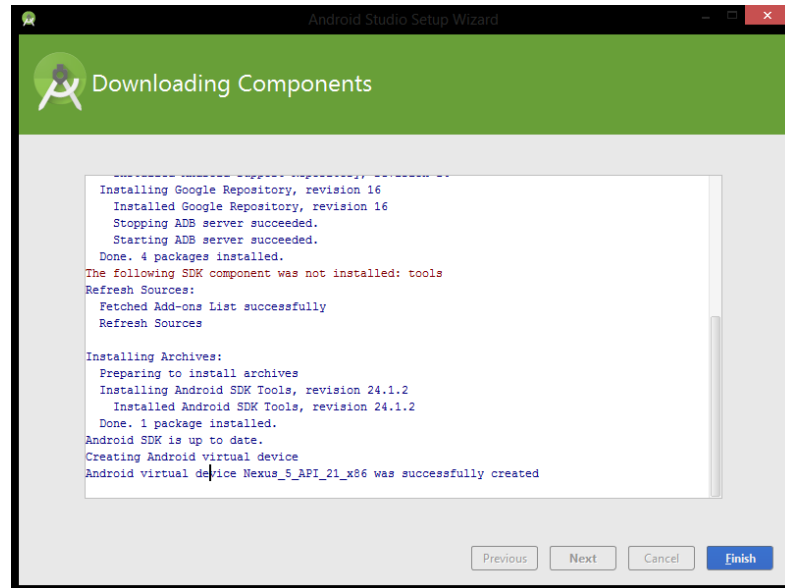
15. As Android Studio starts, the splash screen will appear.



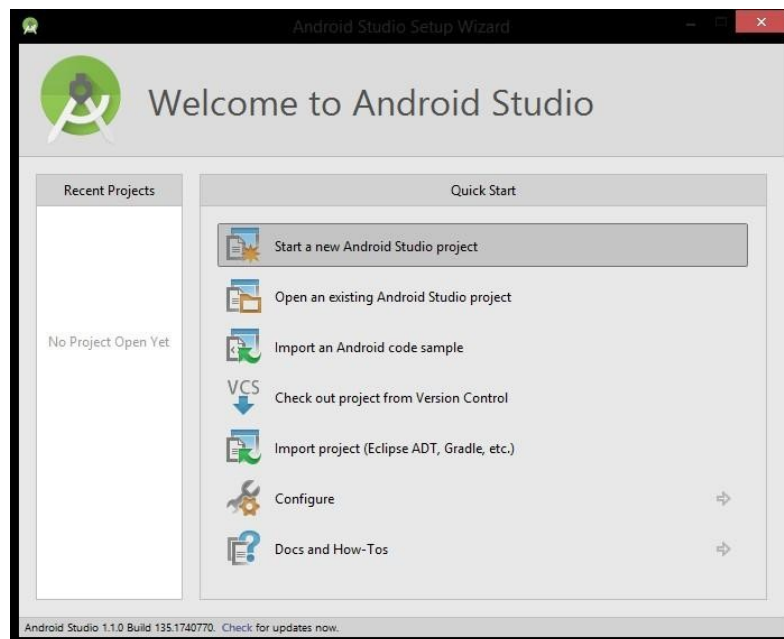
16. After the splash screen you may see some additional setup operations run, such as downloading components.

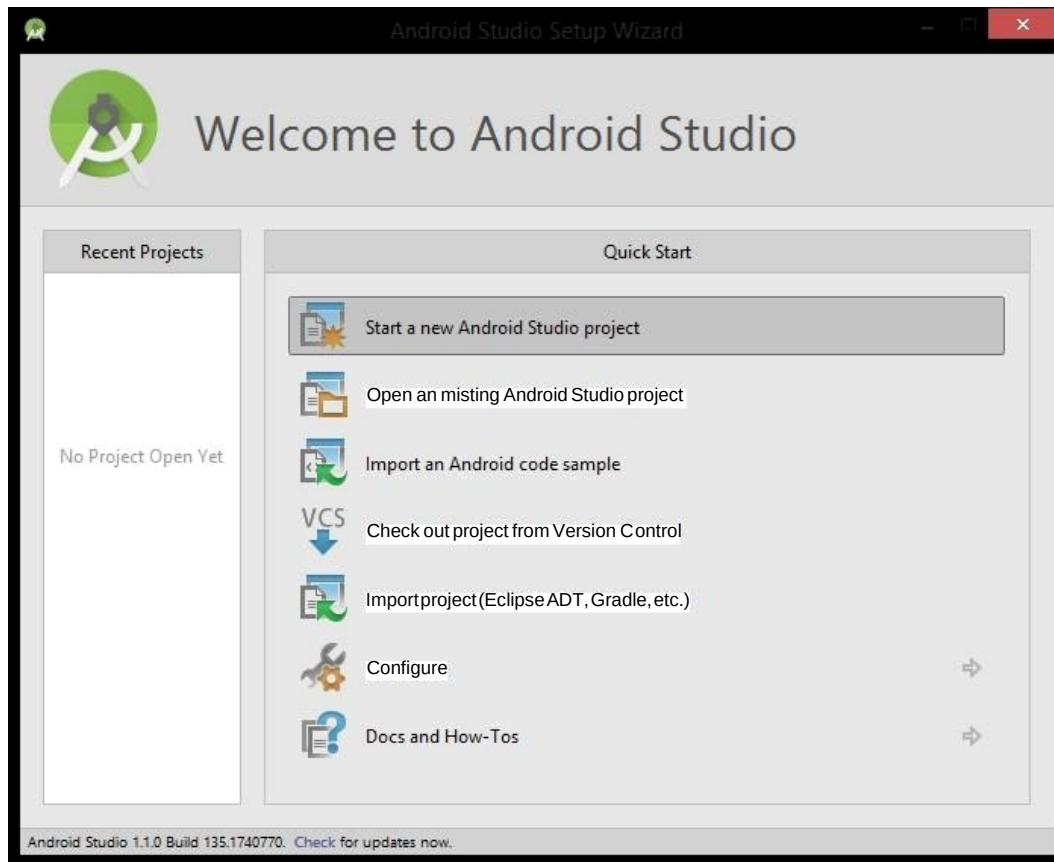


17. Once it is finished, click 'Finish'.



18. Welcome to Android Studio! In the next part we will start our first project.

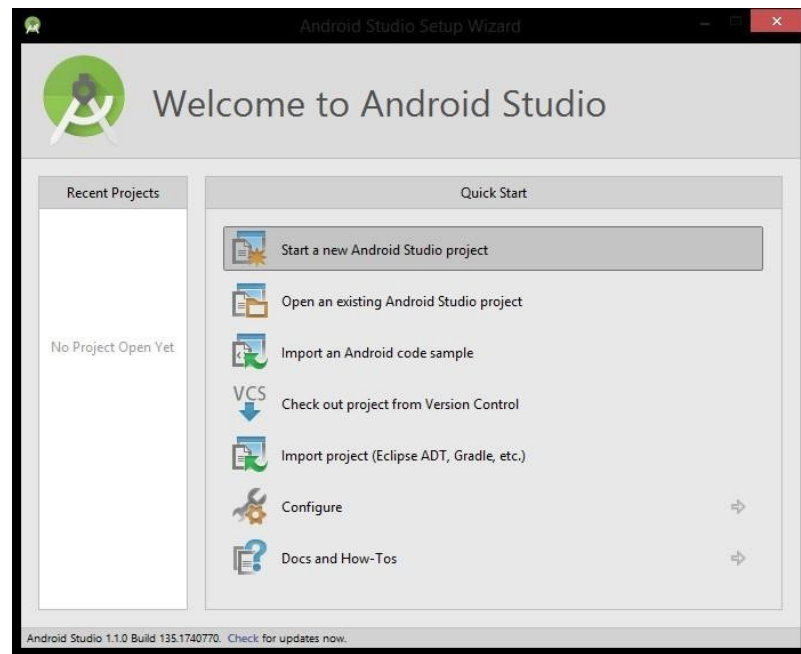




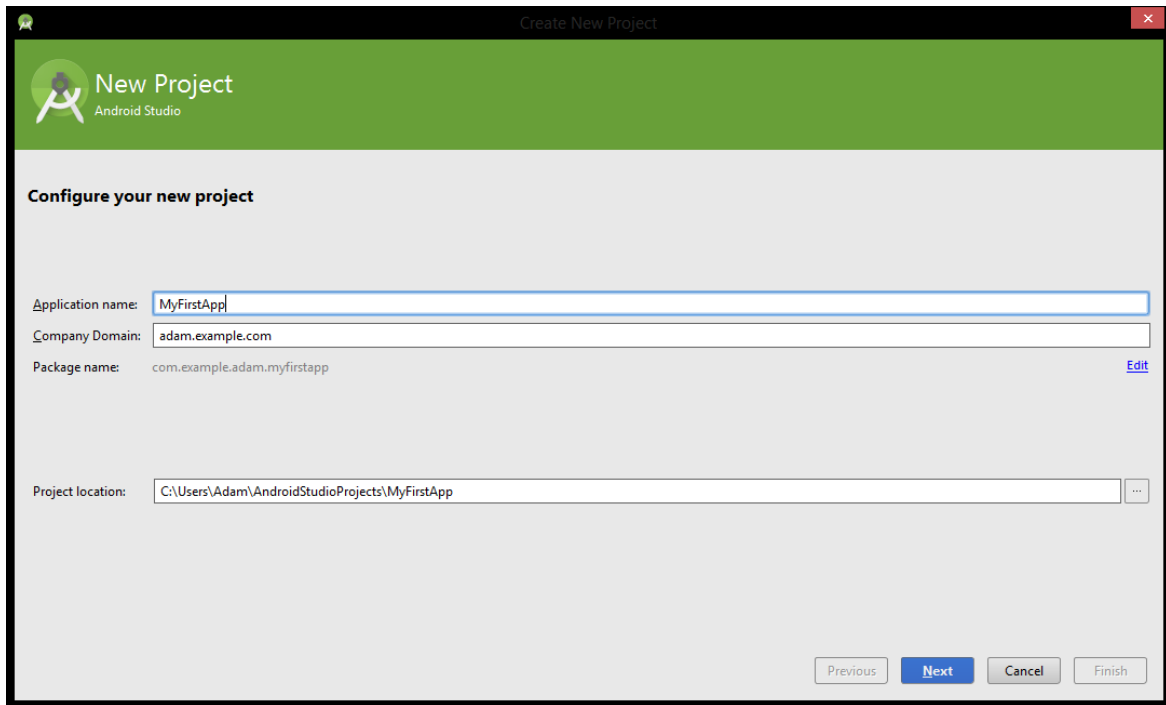
9. Creating A New Project

In this part you will create a simple Android application that displays the words, "Hello World!"

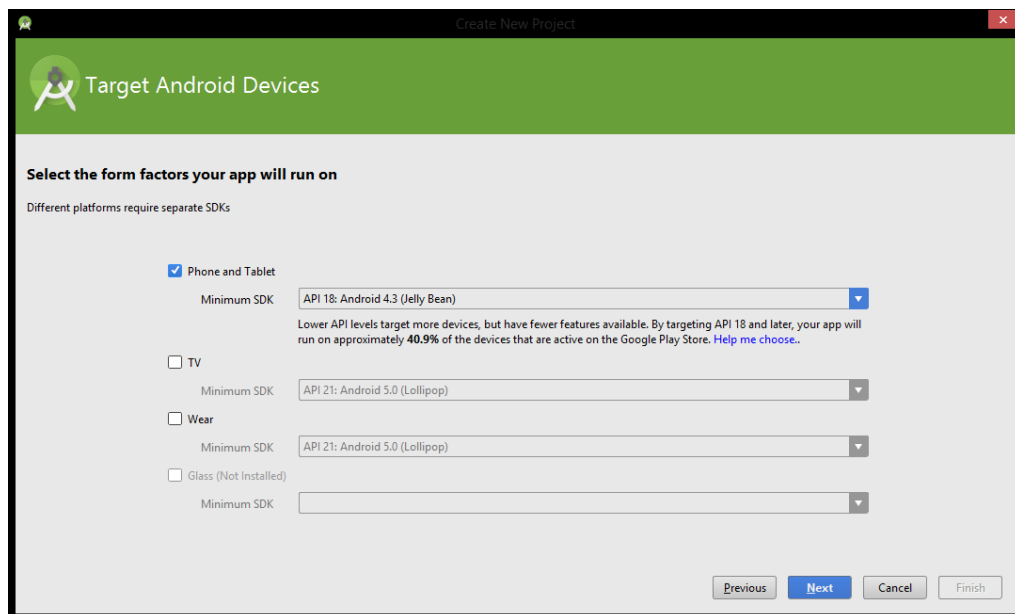
1. At the Welcome Screen, click on 'Start a new Android Studio project'.



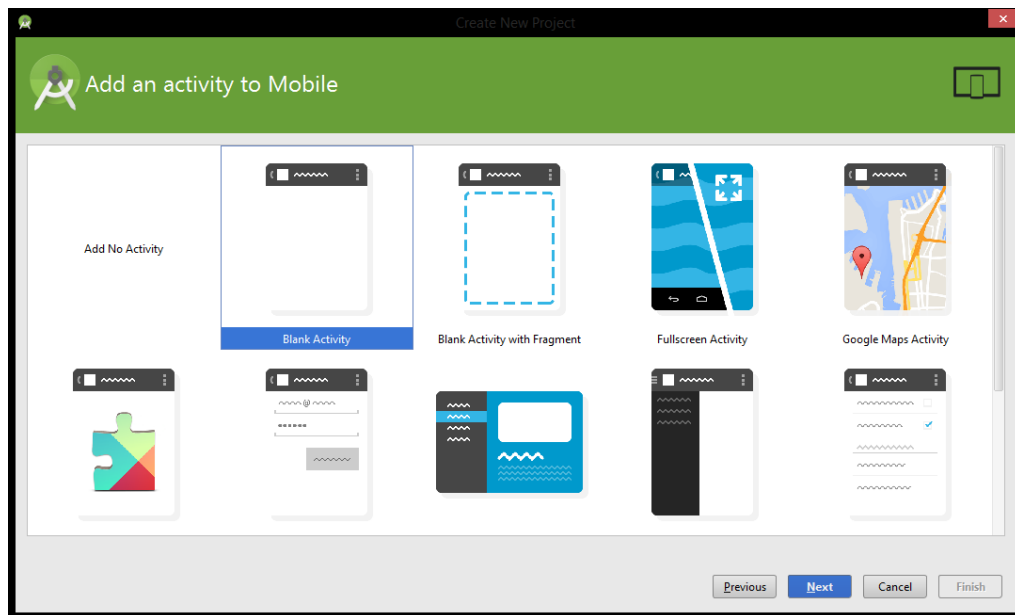
2. Enter the application name 'MyFirstApp' and note where the project is located. The AndroidStudio Projects folder is the default location for new projects.



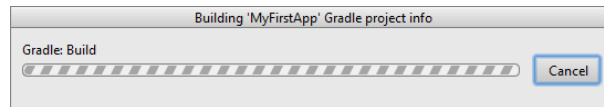
3. Select which devices you would like your app to run on. For now we will be working with 'Phone and Tablet'. Make sure to set the Minimum SDK version to API 18 for this course.



4. Select 'Blank Activity' from the 'Create New Project' dialog box and click 'Next'.



5. In the next window, leave all the settings as default, and then click Finish.
6. Android Studio will now create the project and build it.

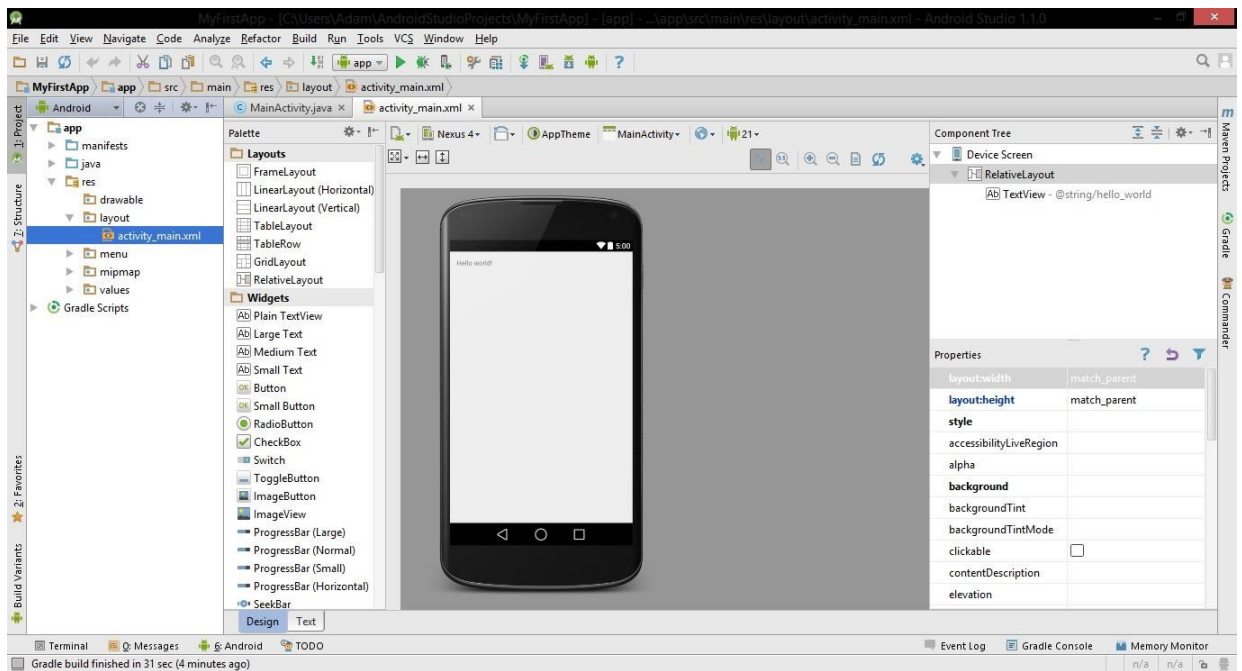


7. You may see a security alert if you are on Windows, click 'Allow access' to continue.



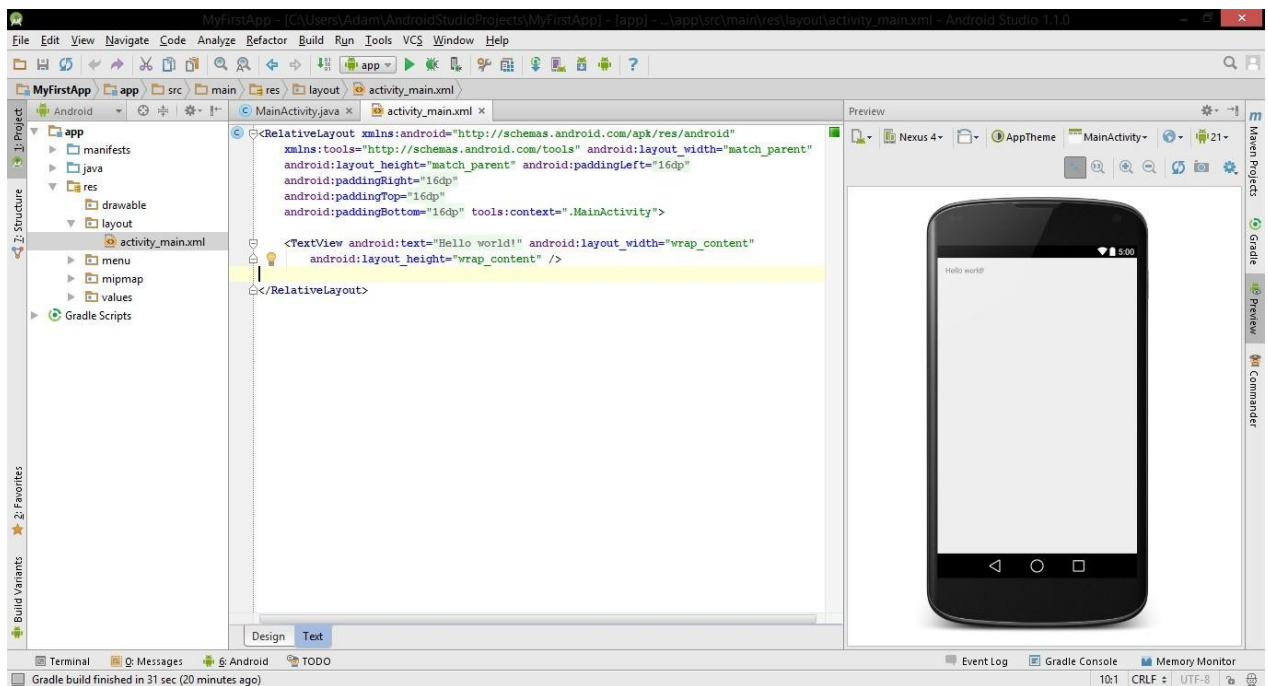
8. Once the Android Studio IDE fully loads, you will see the screen below. If you see text rather than the layout designer make sure the

'Design' tab is selected.



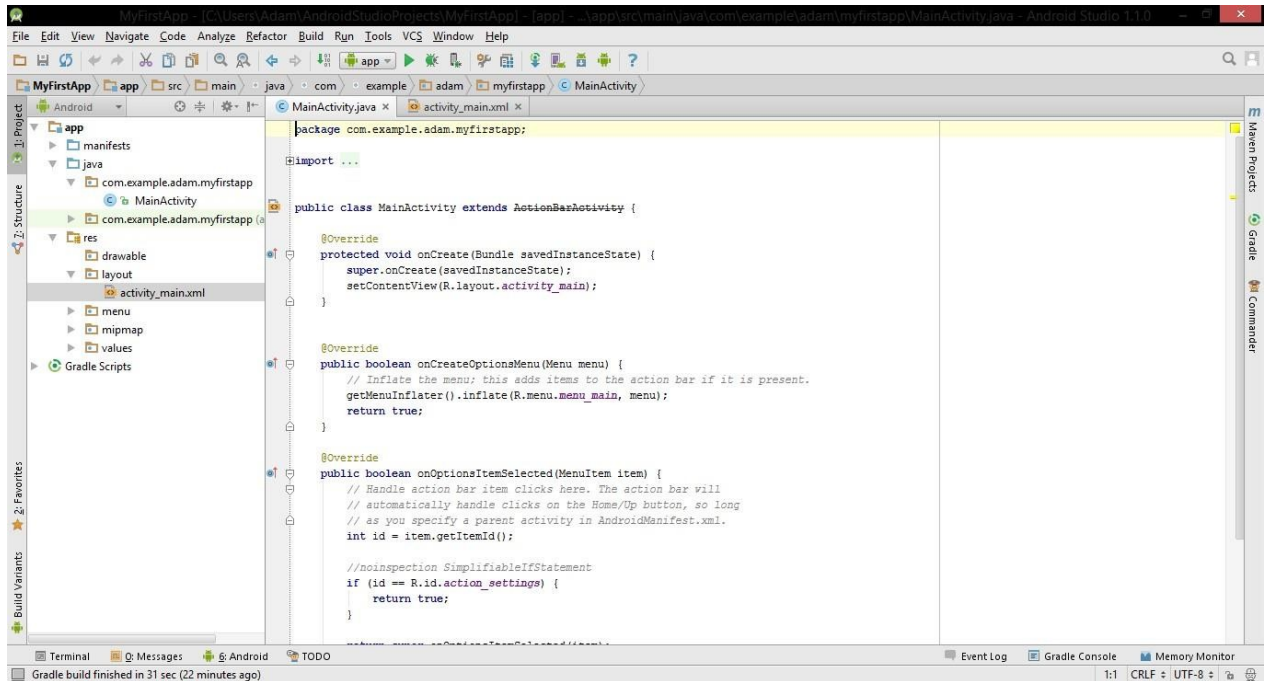
The opening screen is the Design View of the activity_main.xml file. You can already see the words “Hello World!” on the App’s User Interface.

9. If you click on the Text tab you can see the layout file underlying the user interface.



10. To view the backing code for this activity, double click on ‘MainActivity’ inside of the Project directory tree. This file is located in:

'java' > 'com.example.adam.myfirstapp' > 'MainActivity'.

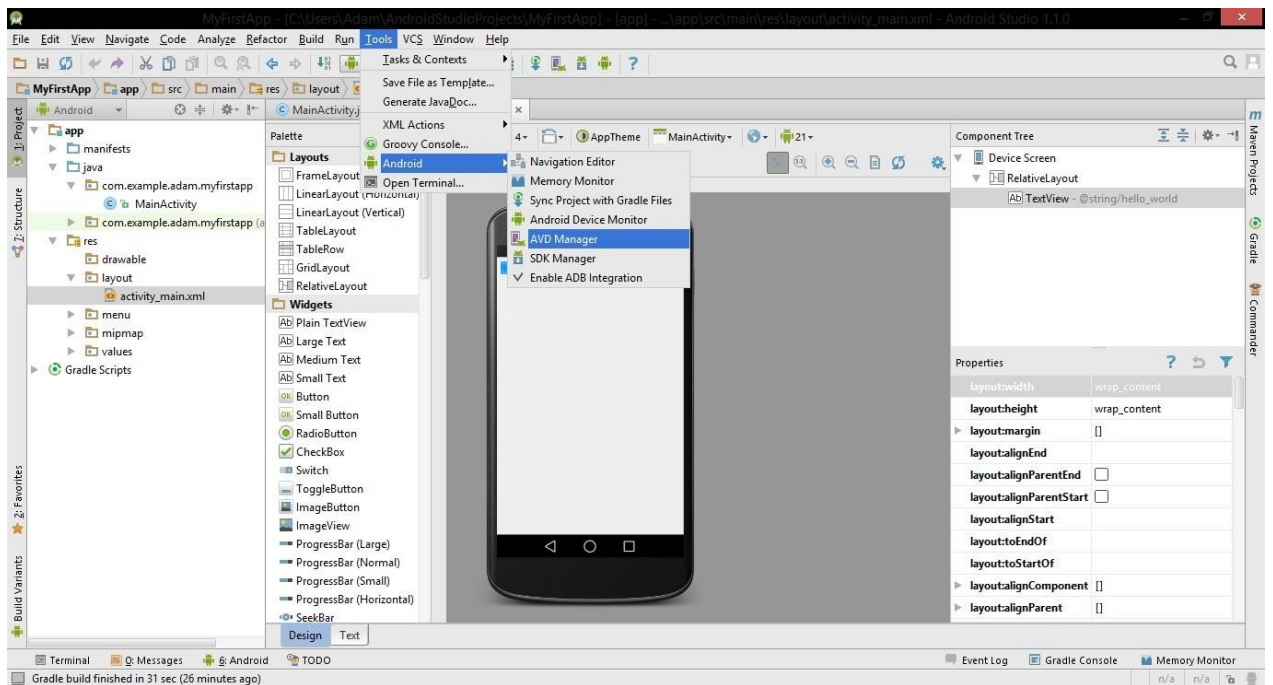


In Part 4 we will show you how to run this app in the Android Emulator.

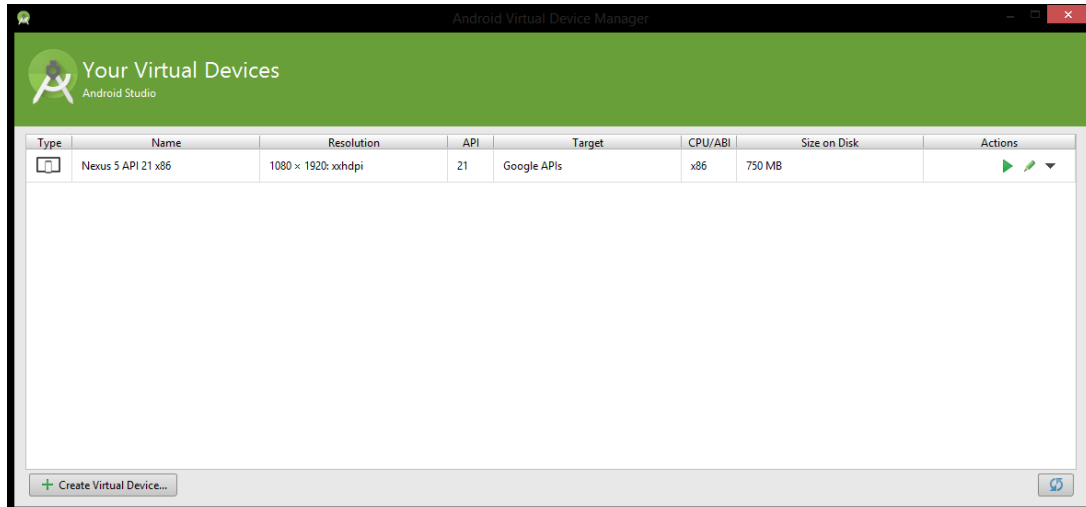
10. Using the Emulator

In this part you will learn how to set up and use the Android Emulator.

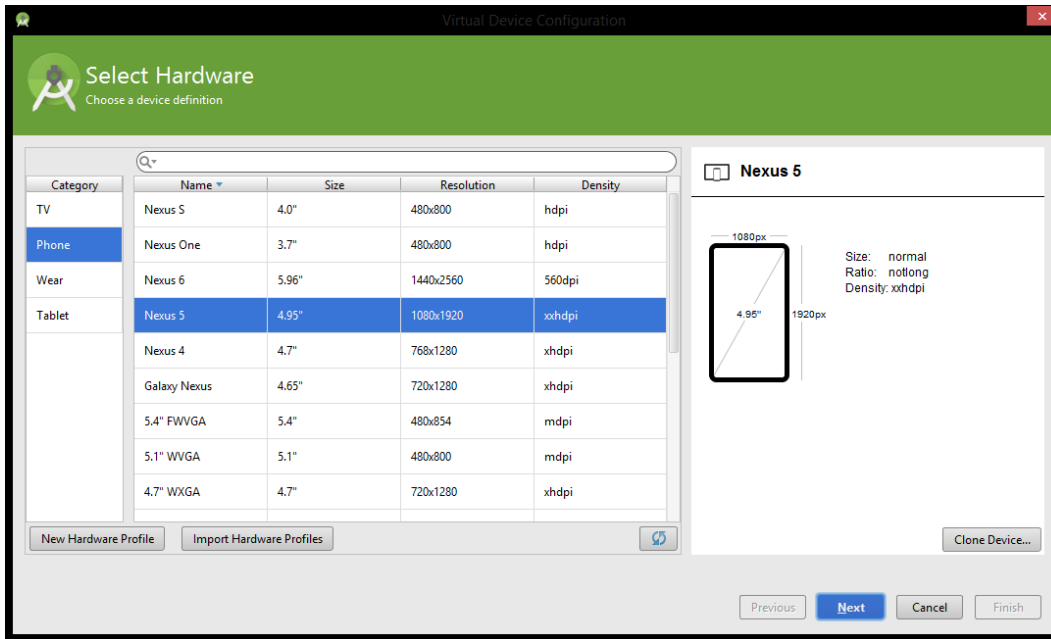
1. First start up the Android Virtual Device Manager. You can do that by selecting Tools > Android > AVD Manager from the Android Studio menu bar.



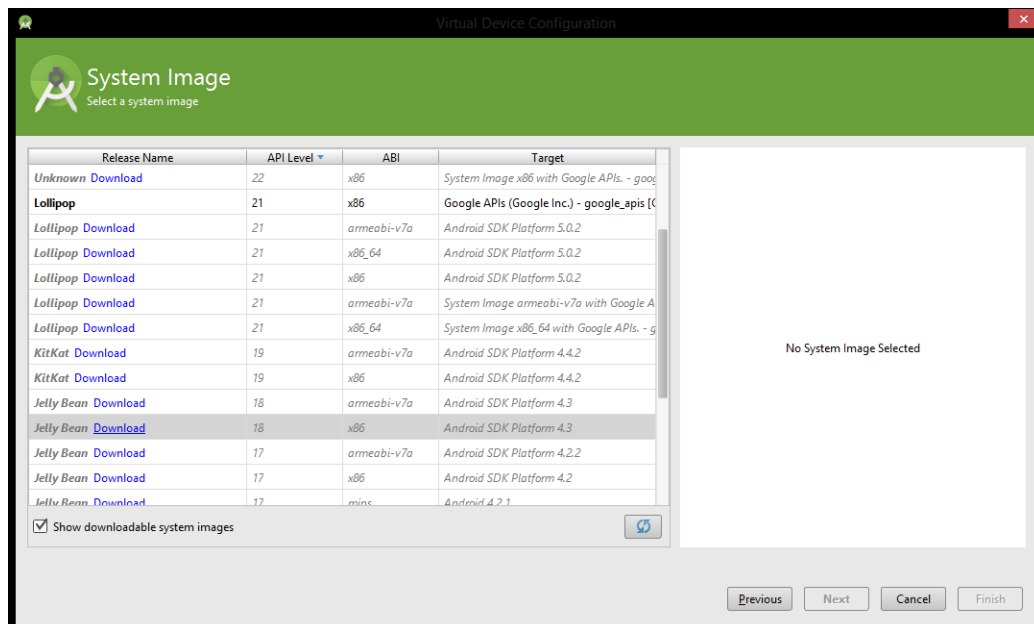
2. A new dialog box will pop up.



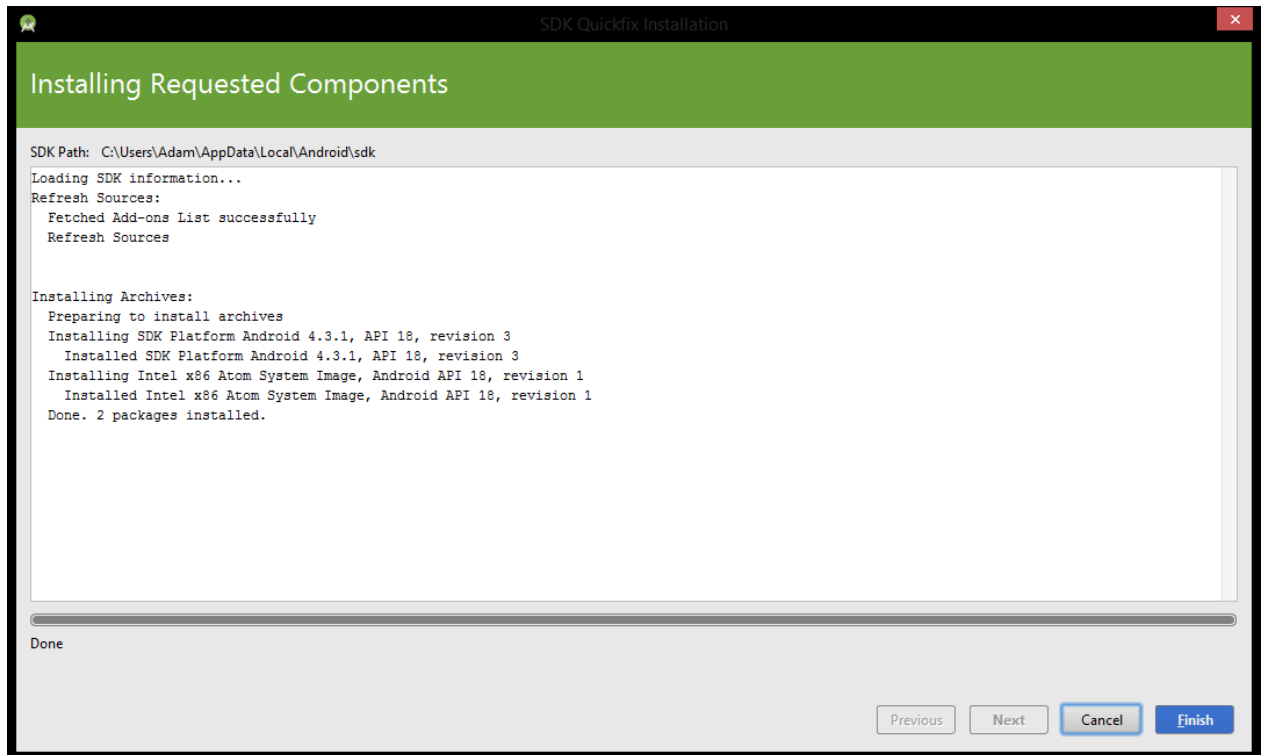
3. Click "Create Virtual Device" to create a new Android Virtual Device (AVD).
4. Another dialog box will pop up displaying various pre-made AVD templates. Select whichever device you would like to emulate and click 'Next'. For example, select the Nexus 5.



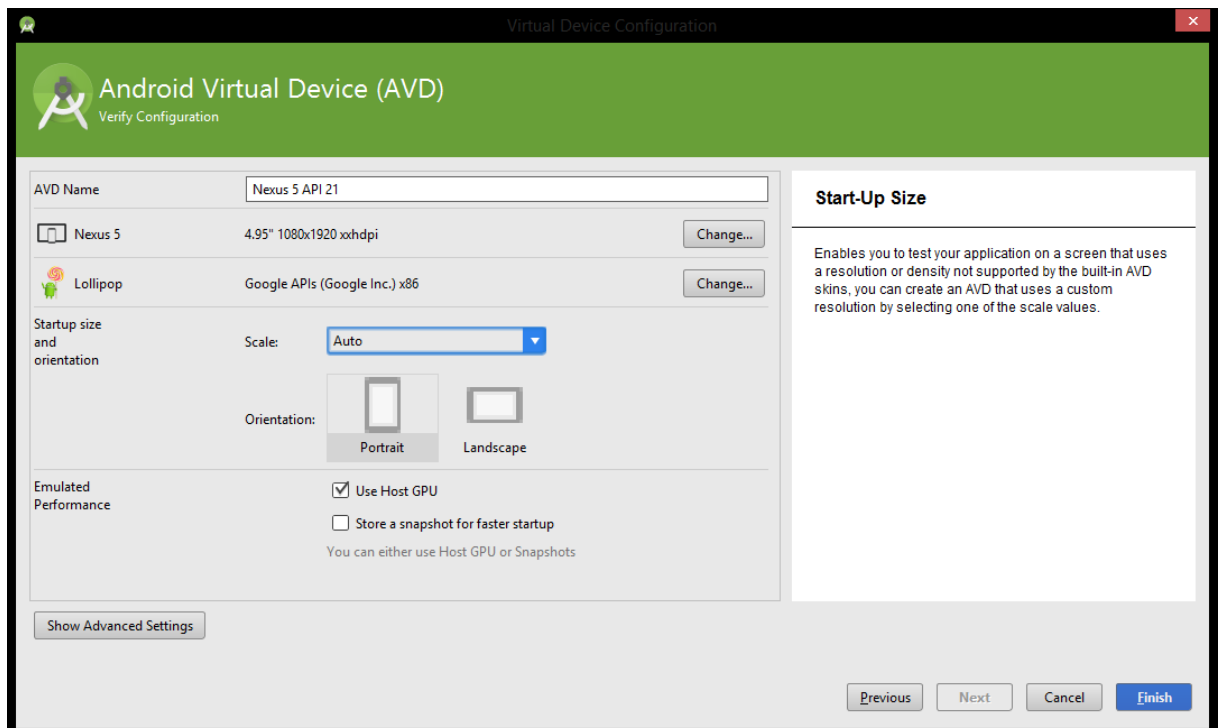
5. Select the appropriate System Image for the virtual machine. To allow users with limited computer memory to participate, all of the class projects will be tested against API level 18. If you haven't downloaded that already, make sure to download it now, by clicking on the "Download" link.



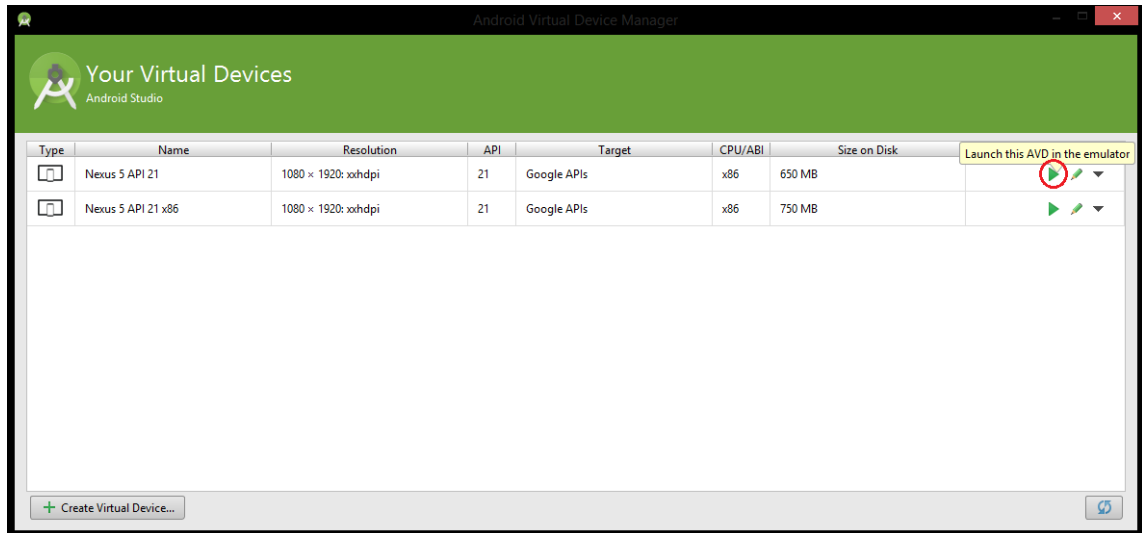
6. Once the Download is finished, click 'Finish'.



7. Click 'Next' once you have returned to the previous screen.
8. You can keep all of the default selections in the next screen and hit 'Finish'.



9. Now click on the green 'Play' icon to start the emulator, after clicking on it you can close the Android Virtual Device Manager.



10. As the emulator starts up, you will see a progress dialog appear in Android Studio.

11. Next, the emulator will appear and start its boot sequence.

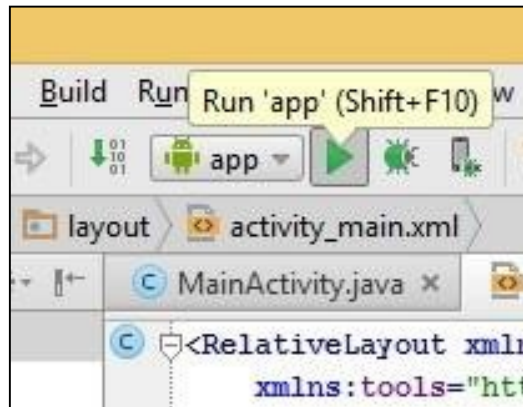


After the device has booted, the emulator will be ready for user interaction.

11. Running Your First App

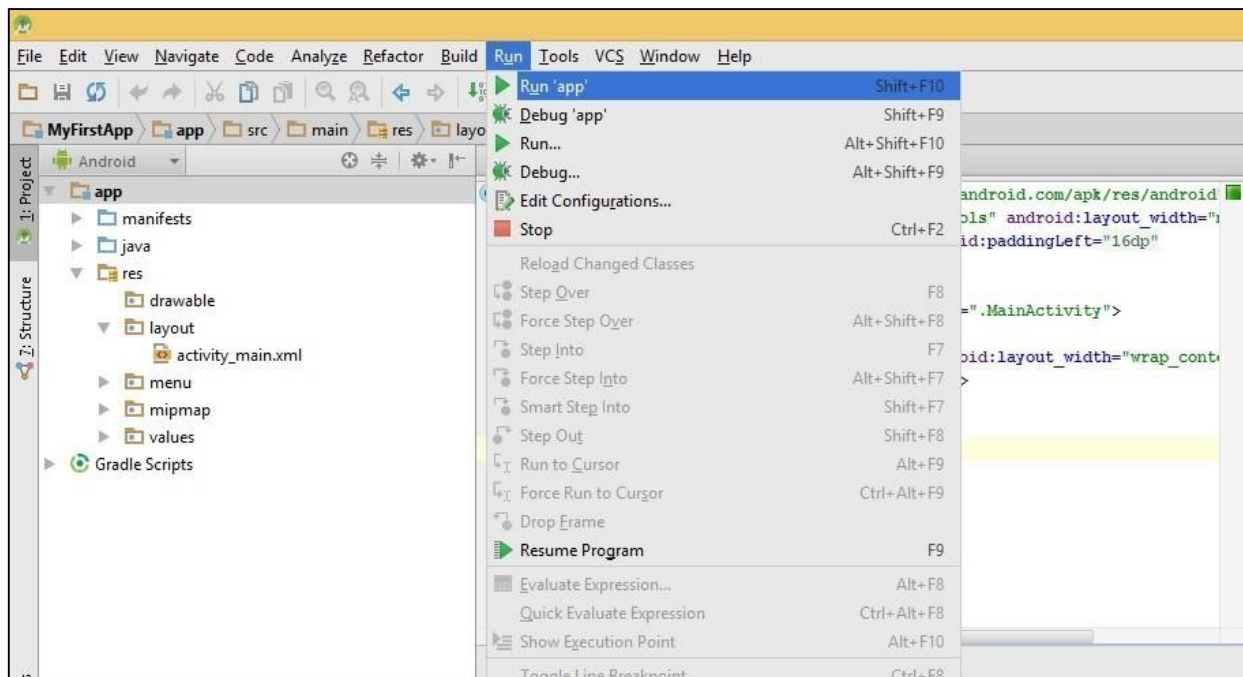
In this part you will learn how to run the application you created in Part 2 in the Android Emulator.

11.1. There are two ways to run the app

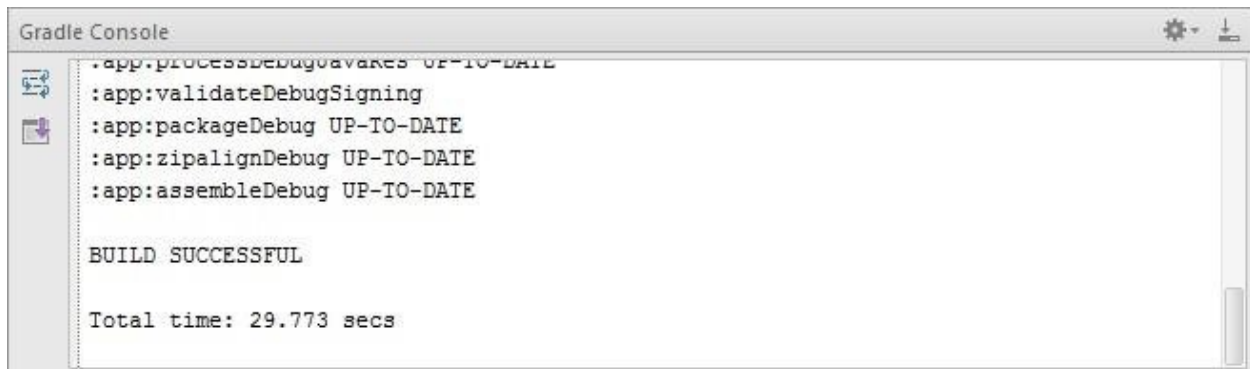


Method 1: Return to Android Studio and simply click on the “Run ‘app’” Button (Shortcut: Windows - Shift + F10)

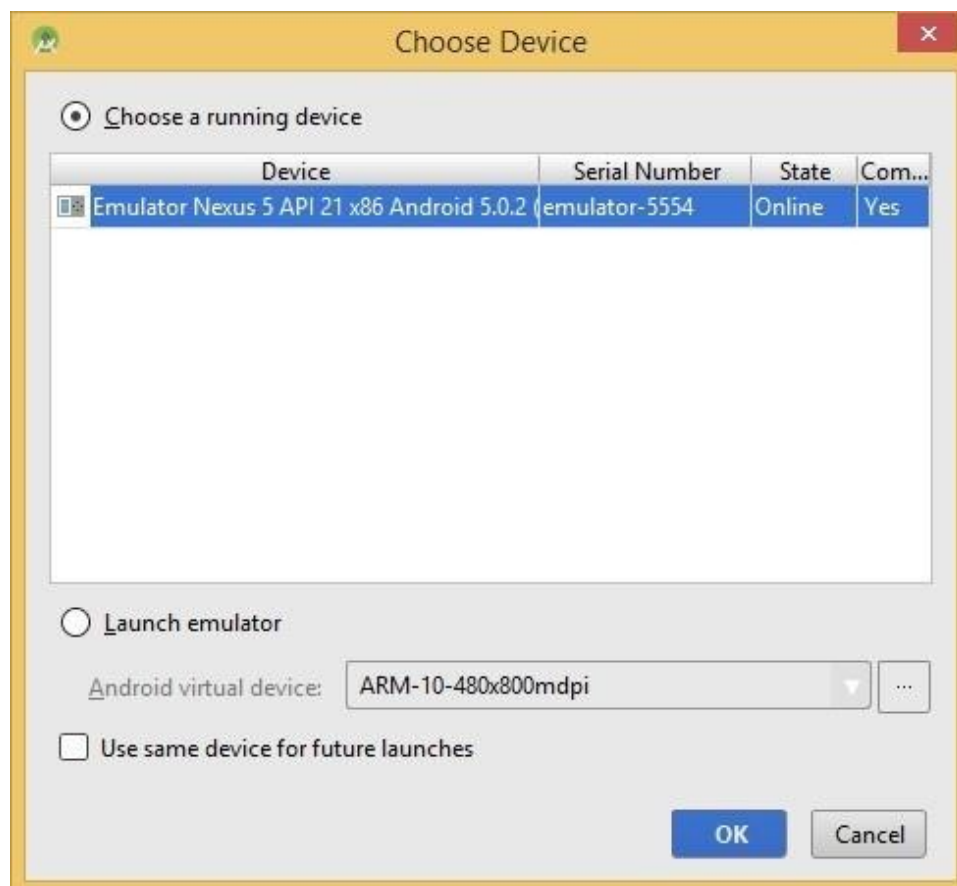
Method 2: Return to Android Studio and select Run > Run ‘app’.



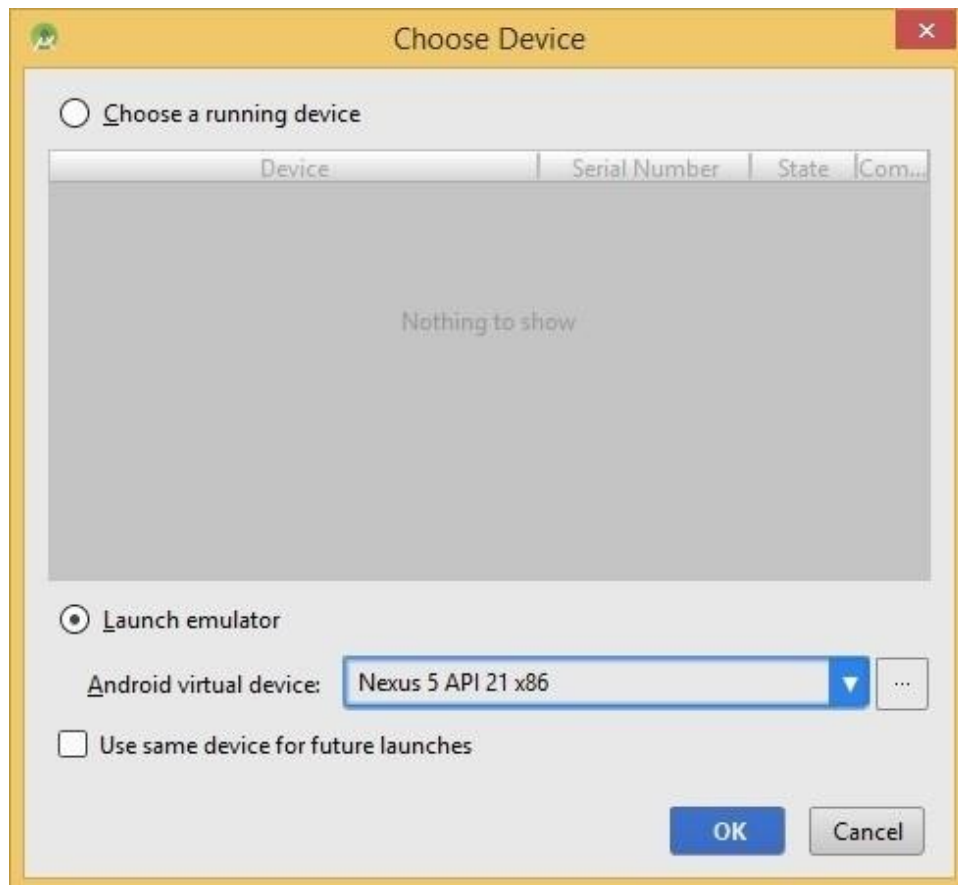
1. In the Gradle Console panel, below the editor window, you will see output indicating that the application is being loaded onto the Android Emulator.



2. After you see the words, BUILD SUCCESSFUL, a dialog box will pop up asking you to choose a device on which to run the app.



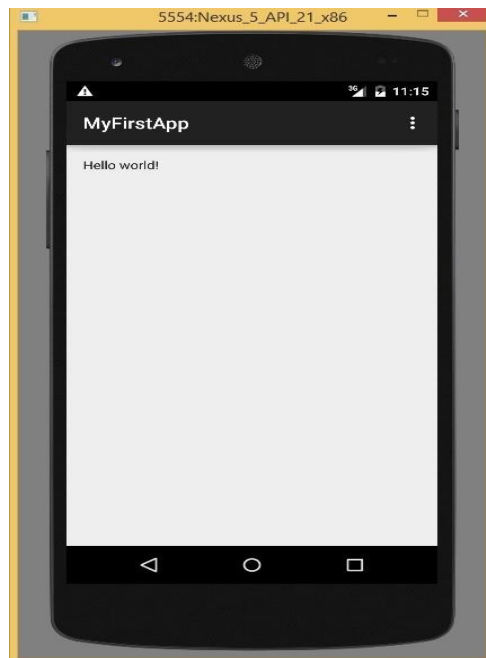
If you do not have a running device, you can choose an AVD to launch in the emulator. The system will start an emulator and run the app in it.



3. Return to your Emulator instance. If necessary, drag the lock icon to unlock your device.



4. You should now see your application, running in the Android Emulator.



12. Practice Tasks

This section will provide information about the practice tasks, which are required to be performed in lab session. Design solutions of problems discussed in each task and place the solution code in a folder specified by your lab instructor.